Communication Networks - Lab 3 Report

Group ID: 13
Student Details:

Name	Family Name	Student ID
Vukašin	Kosijer	s354082
Uroš	Bikić	s352827

1. Introduction

- **Objective:** The objective of this lab is to build a software representation of a point-to-point connection link. We want to simulate signal propagation from a source to a destination by calculating how different distances and various modulation formats affect the transmission quality.
- **Theoretical Background:** Digital signal transmission relies on specific modulation formats, such as BPSK, QPSK, 8QAM, and 16QAM, which transmit 1, 2, 3, and 4 bits per symbol respectively. As the signal propagates, it experiences degradation due to two primary factors: line loss (dependent on length and a loss coefficient) and noise generation. The relationship between the transmitted power, noise power, and line loss determines the digital SNR.
- **Connection to Class:** This lab directly applies theoretical system equations studied in class. It utilizes the formulas for total digital SNR calculation (by subtracting noise and loss from transmitted power in logarithmic units) and translates them into linear units to compute the Bit Error Rate (BER) using $erfc$ function which is adapted for each modulation format.

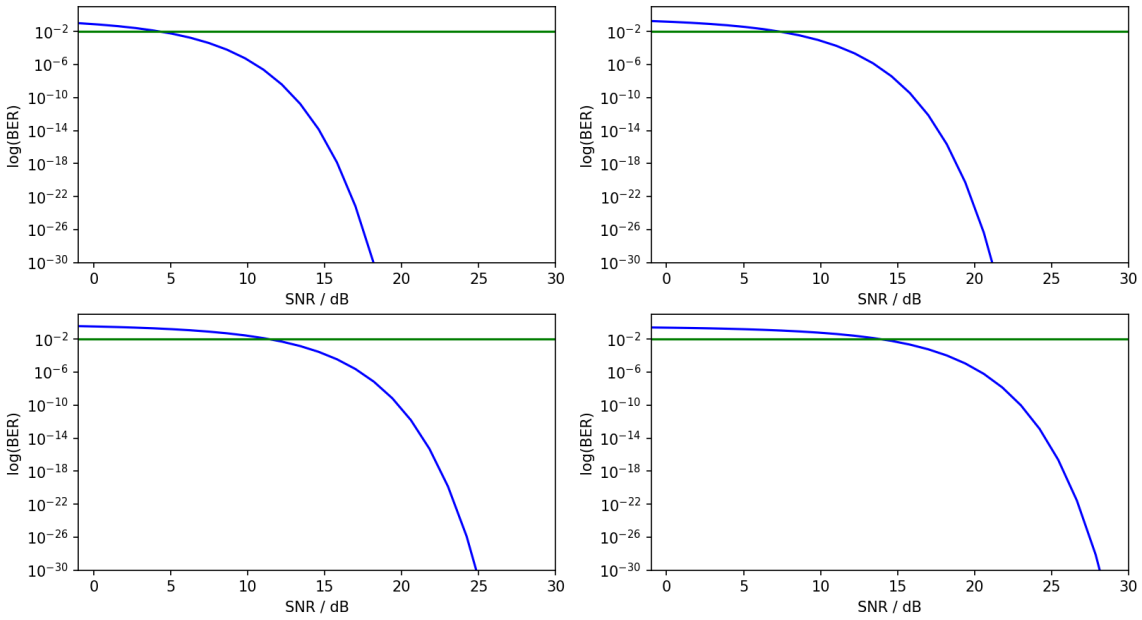
2. Methodology

- **Tools and Libraries:** The simulation was written in Python, utilizing an Object-Oriented Programming approach. To compute the analytical probabilities of error, the `erfc` function was imported from the `scipy.special` library. For data analysis and visualization, numpy and matplotlib.pyplot were used.
- **Key Steps:** We created independent classes for physical components: `Digital_Signal_Information` (managing signal power, noise power, and bits per symbol), `Line` (managing length, loss coefficient, and noise generation), and `PCM` (main container). A method `ber_evaluation()` was created inside the `PCM` class to perform calculations based on $n_{bit,mod}$, returning the specific error probability. Simulations were run across lengths ranging from 10 to 120 km for all modulations using a fixed 1 dB/km α.
- **Justification:** The object-oriented method perfectly maps mathematical concepts to real-world communication blocks. Calculating the powers and losses locally within the class and putting them into the `PCM` container mimics how physical components work.

3. Results and Analysis

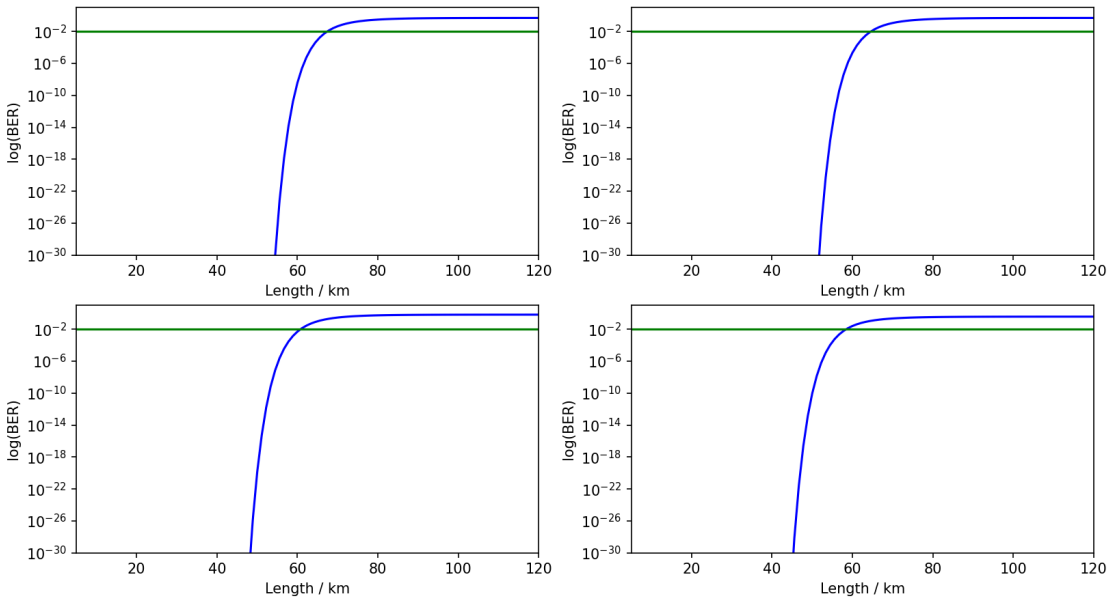
3.1. BER vs. SNR The first set of plots illustrates the relationship between the logarithmic BER and the digital SNR across four different modulation formats: BPSK, QPSK, 8QAM, and 16QAM.

- **Observations:** As expected theoretically, the BER decreases exponentially (shown as a steep downward curve on the logarithmic y-axis) as the SNR increases. To find the minimum required SNR for each modulation format to operate successfully, we locate the exact point where the blue BER curve intersects the horizontal green threshold line.
- **Extracted minimum SNR values:**
    - BPSK (Top-Left): ~4 dB
    - QPSK (Top-Right): ~7 dB
    - 8QAM (Bottom-Left): ~11 dB
    - 16QAM (Bottom-Right): ~15 dB
- **Analysis:** The graphs confirm that higher-order modulation formats are less spectrally robust. Because they pack more bits into a single symbol, it makes them more susceptible to noise. Therefore, they require a significantly higher minimum SNR to achieve the same acceptable error rate as simpler formats like BPSK.



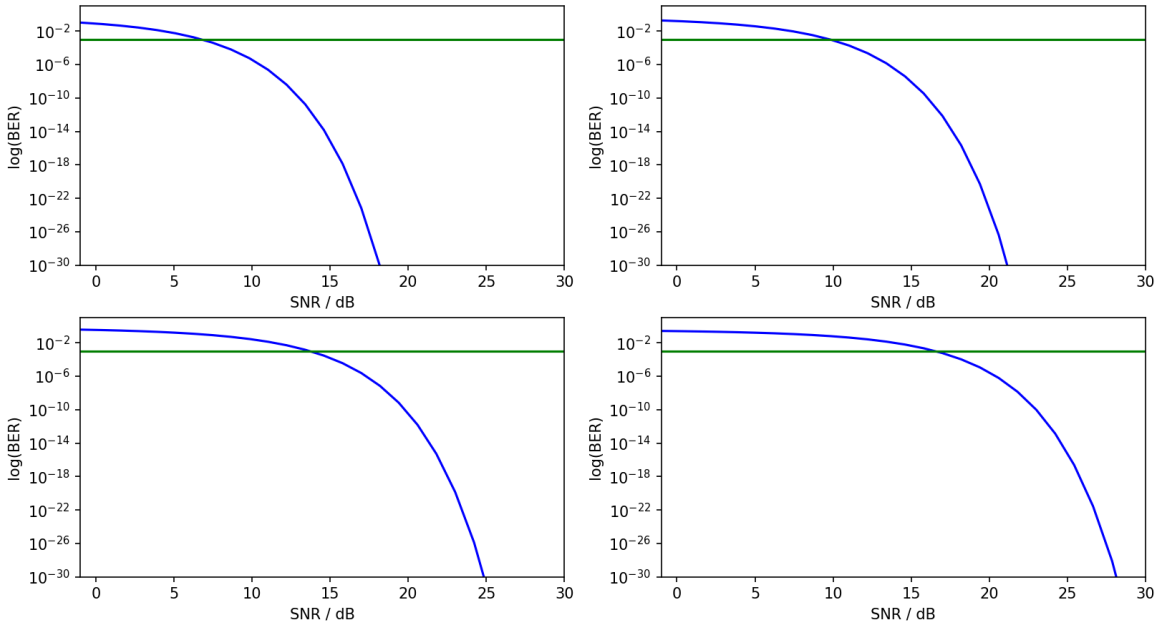
3.2. BER vs. Length The second set of plots demonstrates how the BER degrades as the physical length of the transmission line increases, assuming a standard loss coefficient of 1 dB/km.

- **Observations:** The BER remains extremely low for shorter distances but spikes dramatically once the accumulated attenuation and generated line noise degrade the signal below the critical SNR threshold. The maximum reach is the distance at which the blue curve crosses the green threshold line.
- **Extracted maximum reach values:**
    - BPSK (Top-Left): ~67 km
    - QPSK (Top-Right): ~64 km
    - 8QAM (Bottom-Left): ~60 km
    - 16QAM (Bottom-Right): ~56 km
- **Analysis:** There is a direct inverse relationship between the spectral efficiency of the modulation format and its maximum physical reach. Because 16QAM requires a much higher minimum SNR, it reaches its maximum allowable attenuation much sooner compared to BPSK.



3.3. Impact of a Stricter BER Threshold If the system requirements are set to demand a Maximum Acceptable Bit Error Rate of 10^{-3}, the horizontal green threshold line shifts downward on our plots.

- **New minimum SNR**: Intersecting the curves at a lower point on the y-axis means we must move further to the right on the x-axis. The system now requires a cleaner signal to operate. For example, BPSK's minimum SNR would increase from ~4 dB to ~7 dB, and 16QAM would increase from ~15 dB to ~18 dB.
- **New maximum reach:** Because the system requires a higher minimum SNR, it cannot tolerate as much line attenuation. Intersecting the length plots at a lower y-value shifts the x-intercept to the left. The maximum reach for BPSK would drop from ~67 km to ~63



km, and 16QAM would drop from ~56 km to ~52 km.

3.4. Impact of Varying α The loss coefficient (α) governs how quickly the signal power degrades over distance (Loss = α * Length). Initially, α was set to 1 dB/km.

- **Decreasing α to 0.2 dB/km:** The signal experiences significantly less attenuation per kilometer. Because the signal takes much longer to degrade, the curves on the Length plot would stretch massively to the right. The maximum reach would increase dramatically (roughly scaling up by a factor of nearly 5).
- **Decreasing α to 0.5 dB/km:** The attenuation rate is halved compared to the baseline. This would roughly double the maximum reach of the system across all modulation formats.
- **Increasing α to 2 dB/km:** The line introduces attenuation twice as fast as the baseline. The curves on the plot would become much steeper and shift left, effectively cutting the maximum reach in half.
- **Conclusion:** The physical medium's loss coefficient is the most dominant factor in determining the maximum reach of a point-to-point link. Upgrading to a superior transmission medium allows us to use higher-order modulations effectively.

4. Challenges and Solutions

- **Challenges Encountered:** Managing unit conversions across transmission calculations proved challenging. The laboratory specifications demanded that line loss be calculated with coefficients in dB/km and lengths in meters , while noise generation needed inputs in linear Watts but lengths in meters. Calculating the final digital SNR required converting these values into dBm. Generating readable plots was quite hard as well, due to data's variance.
- **Solutions and Error Sources:** Precision loss due to floating-point representation of extremely small errors was avoided by using logarithmic scales. Specifically, plot limits were manually bounded, restricting the y-axis (representing $\log_{10}(BER)$) to [-30, 0.1] and setting a minimum x-axis limit of -1 to focus only on meaningful positive SNR data.

5. Connection Between Theory and Practice

- **Reinforcing Concepts:** The implementation reinforces the theoretical tradeoff between SE and signal robustness. The simulation proves that while 16QAM carries four times the data of BPSK , it requires significantly higher minimum SNRs to stay below the maximum acceptable BER threshold.
- **Simulated Insights:** By plotting the curves alongside the target horizontal line of $BER_{th} = 10^{-2}$, we can visually identify the critical intersection point. The minimum required SNR and the maximum reach are exactly where the curve crosses this horizontal line.
- **Threshold Variations:** If the system requires higher reliability, by forcing a lower threshold like $BER_{th} = 10^{-3}$, the system demands a substantially higher minimum SNR. Hence, the maximum reach severely drops.
- **Medium Quality Impact:** Adjusting the loss coefficient (α) directly shifts range limits. Switching to an optical fiber coefficient of 0.2 dB/km significantly flattens the loss progression compared to higher values like 2 dB/km, extending the possible maximum reach.

6. Conclusions

- **Key Findings:** The software correctly modelled a point-to-point digital transmission link. We confirmed that as link length increases, the noise and attenuation exponentially degrade the digital SNR, directly causing a spike in the BER. Furthermore, higher-order modulations suffer faster performance degradation over distance, proving that maximum reach is strictly dependent on the chosen digital scheme and the loss coefficient.
- **Future Work:** Possible improvements include testing the system with real-world, non-linear noise or evaluating the impact of varying ADC parameters.